Question Bank

23CST306 - ALGORITHM ANA LYSIS AND DESIGN

Course Type	Course Nature	CA Conduct	System	L	T	P	Credits	CA Total	CA Pass	SEE Total	SEE Pass	Total Pass
Theory	1	End Semester	Mark	3	1	0	4	50	0	100	40	75

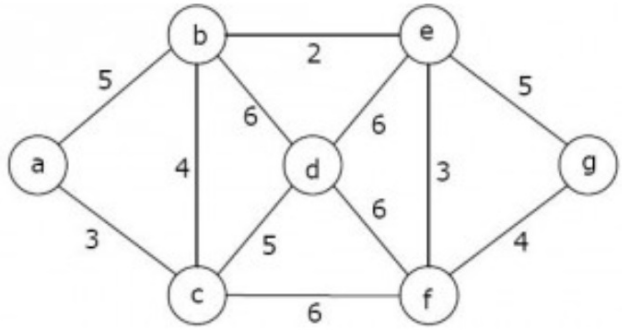
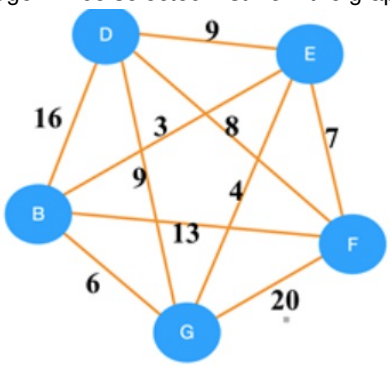
Question Bank Summary

Sect. Part A	Sect. Part B	Easy	Med.	Chall.	Th.	Appli.
36	30	14	30	22	26	40

Part A

#	Unit	Question	COS	Categorized
1	1.1	A. Define an algorithm. List any three characteristics of a good algorithm? (3 marks)	CO1	Challenging - Applying - A
2	1.1	A. Justify how time and space complexity of an algorithm is used for efficiency calculation ?(3 marks)	CO1	Medium - - T
3	1.1	A. State the criteria used for analysing algorithms? (3 marks)	CO1	Easy - - T
4	1.1	A. Which case of complexity is most suitable for algorithm comparison and why? (3 marks)	CO1	Medium - - T
5	1.1	A. Define best case, worst case and average case complexities with f(n) and g(n) functions graph progression? (3 marks)	CO1	Medium - - T
6	1.1	A. Which case complexity is most difficult to compute and why? (3 marks)	CO1	Medium - - T
7	1.2	A. Define Big-Oh (O), Big-Omega (Ω) and Big-Theta (Θ) notations used in algorithm analysis? (3 marks)	CO1	Easy - - T
8	1.2	A. Illustrate the use of asymptotic notations by finding the Big-Oh and Big-Theta of the function **f(n)= $3n^2$ +5n + 7**. (3 marks)	CO1	Challenging - - A
9	1.2	A. Classify and justify the following functions in increasing order of their asymptotic growth rate: **n , n^2 log n , n log n.** (3 marks)	CO1	Medium - - A
10	1.3	A. Calculate the time complexity? for(i=1;i<=n;i++) { for(j=1;j<=n;j++) { x++; }} (3 marks)	CO1	Challenging - - A
11	1.3	Calculate the time complexity? **for(i=1; i<=n; i++) { for(j=1; j<=i; j++) { x++; } }**	CO1	Challenging - Applying - A
12	1.4	State **Master's Theorem** used for solving divide and conquer recurrences.	CO1	Easy - Remembering - T

13	1.4	Summarize the **iteration method** for solving recurrence relations.	CO1	Easy - Understanding - T
14	1.4	Illustrate recurrence relation with an example?	CO1	Easy - Understanding - T
15	2.1	With neat diagrams, explain **LL, RR, LR, and RL rotations** in AVL Trees. Show how these rotations maintain height balance.	CO2	Medium - Understanding - T
16	2.1	Illustrate with an example what happens when the **balance factor exceeds ±1** in an AVL Tree?	CO2	Medium - Understanding - T
17	2.1	State the **time complexity** of search, insertion, and deletion in an AVL Tree.	CO2	Medium - Understanding - T
18	2.1	Why do AVL Trees guarantee **logarithmic height**?	CO2	Medium - Understanding - T
19	2.1	Mention any **two advantages** of AVL Trees.	CO2	Easy - Understanding - T
20	2.1	List the **four types of rotations** used in AVL Trees.	CO2	Medium - Remembering - T
21	2.2	A **social networking application** groups users into communities based on mutual connections. Each user initially belongs to a separate group. The application uses a **Disjoint Set data structure** to manage these groups. a) Identify the **disjoint set operations** used in this scenario. b) Explain how the **Union and Find operations** help in managing user communities.	CO2	Challenging - Applying - A
22	2.2	A **course registration system** groups students who have selected the same elective course. When a student changes the elective, the system updates the groups dynamically. Explain how a **Disjoint Set data structure with Union and Find operations** can be used to efficiently manage the student groups.	CO2	Medium - Applying - A
23	2.2	List the basic operations of a Disjoint set?	CO1	Easy - Remembering - T
24	2.2	**What is the purpose of the Union–Find algorithm?** How does it help in maintaining disjoint sets efficiently?	CO1	Easy - Remembering - T
25	2.3	**Define Depth First Search (DFS).** How does it differ from BFS in terms of traversal strategy?	CO2	Easy - Remembering - T
26	2.3	State the **time complexity of BFS** for a graph represented using an adjacency list?	CO2	Easy - Remembering - T
27	2.3	A **social networking platform** represents users as vertices and friendships as edges.Which graph traversal algorithm (**BFS or DFS**) is suitable to find **all friends at the same level** from a given user? Justify your answer.	CO2	Easy - Understanding - T
28	2.3	A **city transportation map** is modeled as a graph.Which traversal algorithm is preferred to find the **shortest path in terms of number of stops**? Name the algorithm and state its time complexity.	CO2	Medium - Understanding - A
29	2.3	A **project management system** represents tasks as a directed acyclic graph (DAG), where an edge from task *A* to task *B* indicates that *A* must be completed before *B*. Justify how **Topological Sorting** can be used to determine a valid sequence of task execution.	CO2	Challenging - Understanding - A

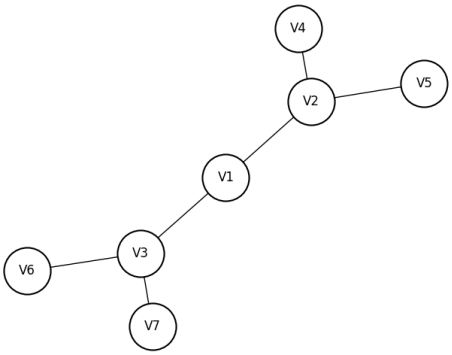
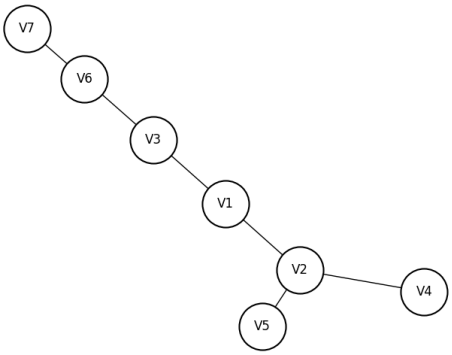
30	2.3	**Define Strongly Connected Components (SCCs)** of a directed graph. How do SCCs differ from connected components in an undirected graph?	CO2	Easy - Understanding - T
31	3.1	Compare the time complexities of Strassen's and ordinary matrix multiplication and analyze which runs faster for large input sizes.	CO3	Medium - Analysing - A
32	3.2	Identify the edges which is NOT added to the minimum spanning tree using Kruskal's algorithm? 	CO3	Easy - Analysing - T
33	3.2	Using Kruskal's algorithm, which edge will be selected first from the graph and what is the time  complexity of Kruskal's algorithm?	CO3	Easy - Analysing - T
34	3.2	Analyze why a greedy algorithm may fail to produce an optimal solution for certain problems.	CO3	Medium - Analysing - A
35	3.2	A city planner wants to connect 6 junctions with minimum road construction cost. If the edges are sorted by weight, identify the first edge that Kruskal's Algorithm will select.	CO3	Medium - Remembering - T
36	3.2	A delivery drone can carry limited weight. Items can be taken in fractions. Apply the Fractional Knapsack Algorithm to decide which item should be selected first.	CO3	Medium - Applying - T

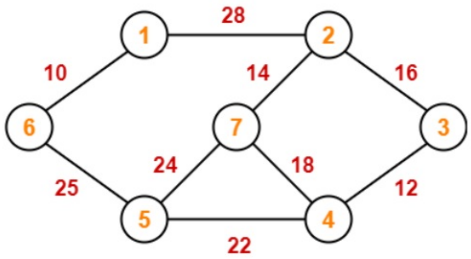
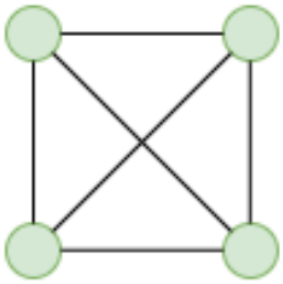
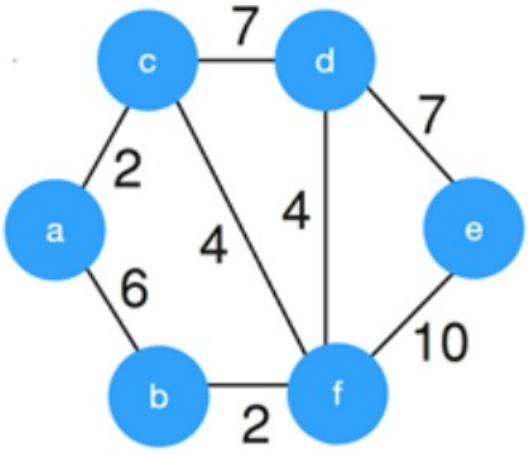
Part B

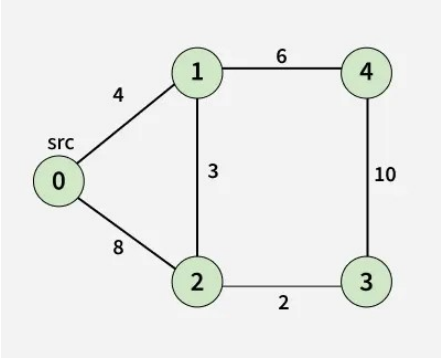
#	Unit	Question	COS	Categorized
1	1.1	A. Give an algorithm that merges three sorted arrays A[1:m],B[1:n] and C[1:k] and yields the output as a single sorted array having m+n+k elements. (9 marks) B. Consider the following algorithm: S=0; for i= 1 to n do S= S + i return i What does this algorithm compute?How many times the basic operation is executed? (5 marks)	CO1,CO1	Medium - - A
2	1.1	A. Analyse the time complexity of the below given algorithm. Algorithm Rsum(a, n) { if (n≤ 0) then return 0.0; else return Rsum(a, n-1)+ a[n]; } (7 marks) B. Analyse the time complexity of the below given algorithm. Algorithm AddMatrix(A,B,n) { for(i=0;i<n;i++) { for(j=0;j<n;j++) { C[i][j]=A[i][j]+B[i][j]; } } } (7 marks)	CO1,CO1	Medium - - A

3	1.2	A. Explain the properties of asymptotic notations used in algorithm analysis. (7 marks) B. Assume the following preposition holds (i) f(n) = O(g(n)) and g(n)=/ O(f(n)) (ii) g(n) = O(h(n)) and h(n)=/ O(g(n)) With proper analysis, prove or disprove the following statements. (1) h(n)=O(f(n)) (2) f(n)+g(n)=O-(h(n)) (3) g(n)+h(n)=/O-(h(n),h(n)) (7 marks)	CO1,CO1	Medium - - T
4	1.3	A. Show the following equalities are correct. 1. $5n^2-6n=\Theta(n^2)$ 2. n! = O (n^n) 3. $2n^2\ 2^n + n \log n = \emptyset(n^2\ 2^n)$ (14 marks)	CO1	Challenging - Evaluating - A
5	1.3	A. Write an algorithm to compute 1+3+5+7+.....+999.Calculate its time complexity using asymptotic notation and justify? (7 marks) B. Prove that for any two functions f(n) and g(n),f(n)=O(g(n)) and f(n)=Ω(g(n)) (7 marks)	CO1,CO1	Challenging - - A
6	1.3	A. A **ride-sharing application** stores trip requests in a list and matches them with available drivers using the following pseudo-code: Algorithm MatchDrivers(requests, drivers) 1. for i = 1 to n: // n = number of requests 2. for j = 1 to m: // m = number of available drivers 3. if Driver[j] is available: 4. assign Driver[j] to Request[i] 5. break 6. end for 7. end for Analyze the **time complexity** and space complexity of the above algorithm in terms of n (number of requests) and m (number of drivers). Suggest a **real-time improvement** to reduce the time complexity if the number of requests and drivers is very large. (14 marks)	CO1	Challenging - - A
7	1.4	A. Write the recursive Fibonacci algorithm and its recurrence relation (7 marks) B. Find the time complexity of the below recurrence relation: T(n)= { 2T(n-1) - if n>0 1 otherwise (7 marks)	CO1,CO1	Medium - - A
8	1.4	A. Solve the recurrence relation using the iteration method. T(n)=T(n−1) + n (7 marks) B. Explain different approaches for solving recurrence relation? (7 marks)	CO1,CO1	Challenging - - A
9	1.4	A. Solve the recurrence using the recursion tree method. T(n)=2T(n/2)+ n (7 marks) B. Find the time complexity of the following recurrence using Master's Theorem: T(n)=3T(n/2) + n (7 marks)	CO1,CO1	Medium - - A
10	1.4	Solve the following recurrences using Master's theorem. a) T(n)=9T(n/3)+n b) T(n)=T(2n/3)+1	CO1	Challenging - Applying - A
11	1.4	A. Solve the following recurrence relation using recursion tree method. T(n)=3T(n/4)+Cn^2 (10 marks) B. Solve the following recurrence relation using master's theorem. T(n)=4T(n/2)+n (4 marks)	CO1,CO1	Challenging - - A
12	1.4	A. Solve the following recurrence relations using master's theorem. T(n)=4T(n/2)+n^3 (7 marks) B. Solve the following recurrence relation using iteration method. T(n)=2T(n/2)+n, T(1)=1 (7 marks)	CO1,CO1	Challenging - Applying - A
13	2.1	A. An AVL tree contains the following keys: **10, 20, 25, 30, 40, 50** a) Construct the AVL tree for the above keys. b) Delete the key **20** from the AVL tree. c) Illustrate the imbalance caused due to deletion and show how it is corrected using suitable rotations. (14 marks)	CO2	Medium - - A

14	2.1	A. A university stores employee IDs in an AVL tree to support fast search operations. The IDs are inserted in the following order: **35, 20, 10, 25, 30, 40, 50** Insert the keys one by one into an AVL tree. • Draw the AVL tree after each insertion • Identify the imbalance cases, if any • Specify the type of rotation(s) used at each step (14 marks)	CO2	Challenging - - A
15	2.1	A. A **digital calendar application** stores events as nodes in an **AVL Tree** to allow fast searching and balanced storage. The events are inserted in the order they are created by users: **March, January, April, February, June, May** a) Insert the events into an AVL Tree **one by one**. b) Draw the tree after each insertion. c) Identify the **imbalance cases** that occur during insertion. d) Specify the **rotations performed** to restore balance. (14 marks)	CO2	Challenging - - A
16	2.1	A. A **library management system** maintains book accession numbers in an **AVL Tree** to ensure fast searching and balanced storage. The current AVL Tree contains the following book IDs: **102, 58, 215, 87, 153, 79, 190** a) Delete the node with key **87** from the AVL Tree. b) Draw the AVL Tree **after deletion**, showing intermediate steps if any rotations are required. c) Identify the **imbalance cases** caused due to deletion. d) Specify the **rotations performed** to restore balance. (14 marks)	CO2	Challenging - - A
17	2.1	A. Explain insertion in an AVL Tree. Construct an AVL Tree for the following sequence of keys and show the necessary rotations at each step:105, 58, 190, 35, 75, 210, 20. (14 marks)	CO2	Challenging - - A
18	2.1	A. An online examination system maintains student registration numbers using an AVL tree. The registration numbers are inserted in the order: **43, 31, 22, 38, 41, 24, 56** Construct the AVL tree by inserting the above keys sequentially. • Show the tree after every insertion • Identify LL, RR, LR, or RL imbalance cases • Explain how balance is restored using appropriate rotations (14 marks)	CO2	Challenging - - A
19	2.2	A. A **network connectivity system** tracks whether computers are connected in a local network. Each computer is initially independent. When a connection is established between two computers, their sets are merged. Using the **Union–Find algorithm**: • Explain how connectivity between two computers is checked • Illustrate the process using a suitable example (7 marks) B. A **road construction department** wants to check whether adding a new road between two cities will create a cycle in the existing road network. Explain how the **Union–Find algorithm** is used to detect cycles in a graph. (7 marks)	CO2,CO2	Medium - - A
20	2.2	A. A **cloud data center** maintains clusters of servers. Servers are merged into the same cluster when they start sharing resources. a) Explain how **Union by Rank** improves the efficiency of merging server clusters. b) Describe the role of **Path Compression** in the Find operation. (7 marks) B. A **gaming platform** maintains teams of players. Players can join teams dynamically, and the platform frequently checks whether two players belong to the same team. Explain how **Disjoint Set operations** are applied in this scenario and analyze their time complexity. (7 marks)	CO2,CO2	Medium - - A

21	2.3	A. Given the following graph, perform BFS traversal starting from vertex V1 and show the order of visiting vertices. Analyze the time complexity of the traversal?  (7 marks) B. A **file system** is organized as a hierarchical structure where folders contain subfolders and files. Explain how **Depth First Search (DFS)** is used to traverse the file system. Illustrate the traversal and analyze its time complexity. (7 marks)	CO2,CO2	Challenging - - A
22	2.3	A. Given a connected graph, **perform DFS traversal** starting from a given vertex. Explain how recursion (or stack) is used and analyze the algorithm.  (7 marks) B. A **social media application** wants to suggest friends based on mutual connections. The user network is represented as a graph. a) Explain how **Breadth First Search (BFS)** can be used for this purpose. b) Illustrate BFS traversal with an example graph. c) Analyze the time complexity of BFS. (7 marks)	CO2,CO2	Medium - - A
23	2.3	A. A **project scheduling system** represents tasks as vertices in a directed graph, where an edge from task *u* to task *v* indicates that *u* must be completed before *v*. a) Define **Topological Sorting** of a directed graph. b) Explain a DFS-based algorithm for topological sorting. c) Perform topological sorting for a given directed acyclic graph and analyze the time complexity of the algorithm. (14 marks)	CO2	Medium - - A
24	2.3	A. A **directed graph** is used to model dependencies among software modules, where each vertex represents a module and a directed edge represents a dependency. a) Define **Strongly Connected Components (SCCs)** of a directed graph. b) Explain **any algorithm** for finding SCCs in a directed graph. c) Illustrate the algorithm with a suitable example graph and show how SCCs are identified. (14 marks)	CO2	Challenging - - A
25	3.1	A company's database contains the ages of seven employees in the following order: [38, 27, 43, 3, 9, 82, 10]. The HR manager wants to prepare an age-wise report, so the ages need to be sorted in ascending order. Analyze the working of 2-way merge sort on this list, showing all steps of splitting and merging, and explain why 2-way merge sort is more efficient than bubble sort for this scenario.	CO3	Medium - Analysing - A

26	3.1	The HR department has two 2×2 matrices representing project scores of employees: Matrix A = [[1, 2], [3, 4]] and Matrix B = [[5, 6], [7, 8]]. Multiply these matrices using Strassen's Algorithm, showing all intermediate steps, and compare the number of multiplications used in Strassen's method with the standard matrix multiplication method. Analyse why Strassen's algorithm is more efficient.	CO3	Challenging - Analysing - A
27	3.2	A. Write the **algorithm** for Kruskal's method to compute a minimum cost spanning tree.(7 marks) B. Construct the minimum spanning tree (MST) for the given graph using Kruskal's Algorithm-  (7 marks)	CO3,CO3	Challenging - - A
28	3.2	A. Draw all possible spanning trees of the graph and list the properties of spanning tree.  (7 marks) B. Consider the given graph and find the weight of the minimum spanning tree using Kruskal's algorithm?  (7 marks)	CO3,CO3	Medium - - A

29	3.2	A delivery company wants to find the shortest routes from its main warehouse to all its branch offices in a city. The road network is represented as a weighted graph, where vertices represent locations and edge weights represent travel time in minutes. Using Dijkstra's Algorithm, determine the shortest travel time from the warehouse (source vertex) to all other branches. Show all steps of the algorithm, including updates of distances and the selection of the next vertex at each stage, and explain why Dijkstra's Algorithm is suitable for this scenario. 	CO3	Medium - Applying - A
30	3.2	A logistics company wants to determine the shortest delivery times from its main warehouse, represented as node 5, to all other distribution centers in a city. The road network is represented as a weighted graph, where vertices represent locations and edge weights represent travel time in minutes. Apply Dijkstra's Algorithm and calculate the shortest travel times from node 5 to all other nodes. Show all steps of the algorithm, including updates of distances and the selection of the next vertex at each stage, and explain why Dijkstra's Algorithm is suitable for finding the quickest routes in this scenario.	CO3	Medium - Applying - A